

Volatility and Volume Effects in European Electricity Spot Markets

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This paper analyses the volatility of wholesale electricity markets for five markets in Europe. Using GARCH models after filtering outliers, significant asymmetric effects and volatility persistence have been documented. Moreover, empirical evidence is provided on a significant relation between volume and volatility which can be both positive or negative depending on the specific market.

1. Introduction and Links to the Literature

In economics and finance, volatility is the simplest and basic criterion to study risks of holding assets with uncertain future values. Moreover, volatility analysis, modelling and forecasting have several applications in risk management and options evaluations.

After the deregulation processes observed in the electricity sector, producers and consumers, and in general all market participants, faced different types of risks. As a consequence, energy derivatives often written on electricity have been formulated and traded, hence it has become important to monitor electricity prices and volatility. Because electricity is a commodity with certain special features not shared with other assets or commodities, it is important to understand its risk properties. This understanding can not only help, for instance, firms to formulate their supply schedules and to evaluate real investments using modern asset pricing techniques but also help consumers to hedge against risk of price spikes or risk associated with physical delivery.¹ Therefore, hedging in power markets deals with extremely volatile prices.

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¹ See Nakamura *et al.* (2006), Regnier (2007), Dahlgren *et al.* (2001) and Zareipour *et al.* (2007), among others.

Electricity markets exhibit very high volatility, it is therefore essential to understand where it comes from. There is not clear evidence of price dynamics in response to demand variations in electricity markets. Walls (1999) and Hadsell (2006) have considered volatility–volume relationships in futures prices together with maturity effects and impacts of new information, while Atkins and Chen (2002), Goto and Karolyi (2004), Karakatsani and Bunn (2004), Kanamura and Ohashi (2007) and Fontana *et al.* (2007) have tried to understand this relationship using spot prices, moving from simple regression analysis to GARCH methodology. But it is still not clear what should be the impact, if any, of total traded volume on price at spot or day-ahead level in European markets. This paper will contribute to the first investigation of European electricity spot volatility, and provide empirical evidence on the volume–volatility relationships at spot level, consider at the same time all features of electricity spot prices without excluding observations, and so release information as in Fontana *et al.* (2007) and Bunn and Gianfreda (2010).

Modelling electricity prices is an important and difficult task because of the existence of *spikes*,² which are one of the electricity price features that will be introduced and explained in the following section. It is standard practice³ to distinguish between two different regimes: a *normal* one, where the spot price is seasonal and mean-reverting; and an *abnormal* one, where the price jumps to high values before reverting, usually in a short time, to normal lower values. Therefore, spikes have been identified by using threshold values and so setting spikes equal to the threshold values. In this way, the spiky magnitude is reduced and the number of observations is preserved.

After filtering the anomalous values, hence focussing on the normal regime, the volatility of German, French, Spanish, Dutch and Italian electricity wholesale prices is examined and explored for its characteristics. GARCH-type models have been tested and specified to find an answer to the important question ‘*Where does electricity volatility come from?*’

The paper is structured as follows. A background on electricity markets with mechanisms of price formulation, price features and data employed are presented in Section 2. The preliminary analysis of stylized facts of spot prices and returns is undertaken in Section 3. The European volatility features are studied in Section 4. The *volume–volatility* relationship is explored in Section 5, where *augmented models* are formulated, estimated and discussed. Section 6 concludes.

² For a detailed overview on this topic see Cartea and Figueroa (2005) and Geman and Roncoroni (2006).

³ See Escribano *et al.* (2002), Kanamura and Ohashi (2004), Huisman and Mahieu (2003), Guirguis and Felder (2004), Weron *et al.* (2004), Mari (2006), Geman and Roncoroni (2006), Fontana *et al.* (2007), Pirino and Renó (2010) and Bunn and Gianfreda (2010).

2. Background on Electricity Markets

2.1. Price Determination

Electricity spot markets are regulated by Power Exchanges or Independent System Operators. Here market participants submit their bids for each hour of the following day, and this is why they are considered *day-ahead markets*. Then, the market is cleared by defining and announcing the equilibrium hourly price for the next day. Physical delivery of electricity is always expected as a natural conclusion of these transactions, hence this market is also considered a *physical market*. Participants can submit and change their bids until the closure of the calling period, when all collected bids are price-ordered and aggregated to create market demand and supply curves for each hour. The bid matching mechanisms ignore capacity constraints of the transmission network and the initial market clearing price is determined by the intersection of the aggregated demand and supply curves. If there is no intersection, then different mechanisms are adopted in each market to obtain the *market clearing price*.

2.2. Stylized Facts of Electricity Spot Prices

Electricity prices display interesting features at daily level: *mean-reversion*, *seasonality*, *volatility clustering*, *extreme values* (or *peaks*) and *inverse leverage effect*. The *mean-reversion* is the behaviour of prices to tend to a long-term equilibrium level. A reasonable explanation could lie on the cyclical and mean reverting process of weather conditions, industrial activities or economic cycles. The *seasonality* can be explored in three components: daily, weekly and monthly. The principal seasonal behaviour can be explained by considering day-and-night times, business activities, calendar seasons and general weather conditions which heavily influence the electricity demand. There is not a good definition of *volatility* and it is usually meant as variability of prices. Therefore, one can use the *variance* as measure of volatility to indicate the dispersion of prices around their mean level. The electricity prices are found to be highly volatile showing characteristics as *volatility clustering*, periods in which prices move less are followed by periods in which they move more. Another feature is the *leverage effect*, that is volatility reacts differently to shocks: on financial markets, it has been observed that a negative shock will increase the volatility more than a positive one. But Knittel and Roberts (2005) discovered the *inverse leverage effect*: they found that electricity price volatility has an inverse reaction to shocks and it tends to rise when positive shocks occur. *Extreme values* or *peaks* are results of abnormally large variations in price. The peculiarity of evolution of jumps is that

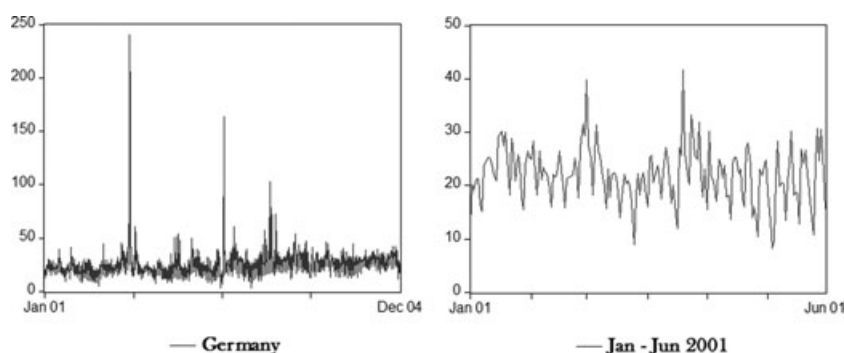


Figure 1. Stylized Features on German Market

the price does not stay on the new level to which it moves, but it tends to revert rapidly to the previous level. These jumps could be caused by factors affecting the demand curve, as weather conditions, or outages and transmission failures which impact on the supply curve instead.

2.3. Data Set Employed

Data sets are composed of 24-hour prices and volumes expressed in €/MWh and in MW, respectively. Average daily prices and volumes have been computed as in Lucia and Schwartz (2002). The employed time series have the following length: 1665 observations for Germany, 2828 for France, 1607 for Spain, 1673 for The Netherlands and finally 1948 for Italy.⁴

The dynamics of German electricity prices is shown in Figure 1, where it is possible to observe mean-reversion, spikes and volatility clustering on the left, whereas seasonality is more visible on the right; similarly for the other markets.

3. Preliminary Investigation of European Spot Prices and Returns

To the author's knowledge, there is not a complete empirical analysis based on stylized facts of European Electricity price volatility. Therefore, the presence of mean-reversion, seasonality, clustered and asymmetric volatility has to be investigated.

⁴ To be precise, German, Spanish and Dutch time series begin on 1 January 2001 and end on 31 July 2005, whereas French series starts on 27 November 2001 ending on 24 August 2009. Finally, Italian series starts on 1 April 2004 and ends on 31 July 2009.

The understanding of spot price dynamics is important for several reasons: first of all, proper modelling is essential to monitor market mechanisms and control the deregulation process; secondly, a correct model specification helps in forecasting prices and in scheduling production plants capacity; thirdly, problems of pricing and hedging energy derivatives are strictly connected to the underlying electricity price process.

The analysis of these price dynamics has been executed on the entire sample (where AS means ‘all sample’) and on time series cut at three threshold values,⁵ which are respectively one, two and three times the sample standard deviations. Logarithmic returns have been computed as $r_t = \log p_t - \log p_{t-1}$, where p_t represents the daily average price at day t of the 24-hour prices observed on day t . From the descriptive statistics reported in Table 1, it is possible to observe that the existence of very high maxima and very low minima prices, on the entire sample, determines large standard deviations, indicating a strong variability in these dynamics. Moreover, spikes heavily influence skewness and kurtosis of distributions which seem to be normal, apart for Italy. When thresholds on prices are introduced, the influence of abnormally large observations is reduced towards the normality assumption. Considering returns, it is possible to notice that while the mean values are positive, the median ones are negative. All return series are extremely volatile with daily standard deviations moving in a range from 22% to 44% of the mean. They exhibit positive skewness due to huge upward movements in prices and so their distribution is also unlikely to be normal.

Among stylized facts of electricity prices, high volatility represents the most important and evident feature. Comparing volatility in electricity markets with those observed in other financial markets, it can be observed that the former is very high, up to 50%, against a maximum of 4% of the mean observed for stocks, crude oil or natural gas, or again compared with stock indexes with a volatility less than 1.5% and even less so for treasury bills (see Weron, 2006). Looking at possible causes of this feature, previous studies on electricity volatility referred to the influences of congestions and technologies that determine the zonal prices in the Italian market (Gianfreda and Grossi, 2009); or to changes in the market structure such as the introduction of certificates on energy efficiency and the liberalization of the demand together with the supply side (Petrella and Sapio, 2009); or finally to transmissions of volatility shocks across European markets (Bunn and Gianfreda, 2010). But, still, there are not answers on the properties of electricity spot volatility and on its possible causes. Then, this empirical investigation is firstly focussed on testing if the *stylized facts* of prices are

⁵ Accounting for spikes, the approach of separating spikes from the normal dynamics (as suggested in literature by Fontana *et al.*, 2007; Weron *et al.*, 2004; Duffie *et al.*, 1998) is used setting jumps equal to the threshold values, $J = (50, 70, 90)$ €, when they exceed the thresholds, that is reducing the magnitude of spikes without excluding observations.

Table 1: Descriptive Statistics for European Electricity Prices and Returns

		Mean	Median	Maximum	Minimum	SD	Skewness	Kurtosis
Germany	Prices	AS	27.950	26.500	240.260	3.120	12.884	65.307
		50	27.328	26.500	50.000	3.120	9.508	2.888
		70	27.343	26.380	69.220	3.120	9.871	4.092
	Returns	AS	27.524	26.420	89.420	3.120	10.306	5.553
		50	0.001	-0.037	2.369	-1.963	0.339	7.469
		70	40.574	33.918	314.269	4.929	23.559	19.222
France	Prices	AS	34.545	33.655	50.000	0.000	12.062	2.147
		50	34.354	32.005	69.965	4.929	13.726	2.942
		70	37.617	33.159	89.947	4.929	17.603	3.299
	Returns	AS	0.000	-0.031	2.776	-1.918	0.305	8.103
		50	32.521	31.435	103.758	5.468	11.509	5.678
		70	31.936	31.435	50.000	5.468	9.934	2.006
Spain	Prices	AS	32.038	31.289	69.808	5.468	10.462	2.813
		50	32.436	31.430	88.456	5.468	11.262	4.544
		70	0.001	-0.014	1.185	-1.153	0.218	5.428
	Returns	AS	36.353	30.350	660.344	2.050	32.668	161.916
		50	31.656	30.350	50.000	2.050	10.675	2.251
		70	31.202	29.628	69.754	2.050	11.375	3.824
The Netherlands	Prices	AS	32.156	29.895	88.486	2.050	13.020	5.085
		50	0.001	-0.030	3.539	-2.533	0.440	9.617
		70	69.039	68.696	142.596	1.211	17.726	3.038
Italy	Returns	AS	48.815	50.000	50.000	1.211	4.063	33.072
		50	55.473	56.273	69.976	1.211	10.542	4.024
		70	65.124	66.554	89.990	1.211	14.634	2.791
		AS	0.000	-0.005	3.567	-3.286	0.244	55.597

Table 2: ARCH LM Test with Six Lagged Residuals

	Germany	France	Spain	Netherlands	Italy
ARCH LM test statistics	51.925	159.276	123.716	85.565	687.885
<i>p</i> -value	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Number of significant lagged $\hat{\varepsilon}^2$	2	2	2	3	5

also reproduced by volatilities, and secondly on a possible explanation for this electricity price variability.

4. Analysis of European Spot Volatility

To account for *volatility clustering* and asymmetries, also called the *inverse leverage effects*, generalized autoregressive conditional heteroskedastic models, GARCH (Bollerslev, 1986), have been considered.

To perform a uniform analysis on markets with different autocorrelation structures, the simplest autoregressive model has been adapted for the conditional mean equations, using time series adjusted for seasonality. Residuals have been tested for the existence of *volatility clustering* implementing Engle (1982) on squared lagged residuals and results, presented in Table 2, confirm the time-varying volatility. Moreover, the *negative bias test* of Engle and Ng (1993) has been used to detect asymmetries in volatility.

In details, residuals ($\hat{\varepsilon}_t$) of the autoregressive models estimated on each market have been used to compute a dummy accounting for negative signs, that is $N_t^i = 1$ if $\hat{\varepsilon}_t^i < 0$ or zero otherwise for $i = \text{Germany, France, Spain, Netherlands and Italy}$. And finally, using the regression $\hat{\varepsilon}_t^2 = \phi + \phi_1 N_{t-1} + \eta_t$ where η_t is an i.i.d. error term, it is possible to understand if the conditional variance is affected by negative shocks when the estimates of ϕ_1 are statistically significant. Because all ϕ_1 coefficients are found to be statistically significant⁶ in all markets with respect to the critical values, there is an empirical evidence on asymmetries in the dynamics of electricity price volatility.

As in Hadsell (2006), but considering in addition *mean-reversion* and the *inverse leverage effect*, the following models have been tested to verify the appropriateness of including asymmetric and clustered volatility in European markets. The following two models are proposed and compared: the first model, (M1), is a combination of a basic autoregressive model in Equation (1) and a simple GARCH(1,1) for the conditional variance

⁶ Parameter estimates are equal to 0.093, 0.077, 0.031, 0.076 and 0.067, respectively, in Germany, France, Spain, The Netherlands and Italy.

(Equation 2); and the second model, (M2), is a combination of the same basic model but with a different specification for the variance equation which is formulated according to an Exponential GARCH⁷ (Equation 3):

$$(1) \quad r_t = \mu + \delta r_{t-1} + \sum_{i=1}^6 c_i D_i + \varepsilon_t$$

$$(2) \quad h_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1}^2$$

$$(3) \quad \log(h_t^2) = \omega' + \alpha' \left(\frac{|\varepsilon_{t-1}|}{h_{t-1}} - \sqrt{\frac{2}{\pi}} \right) + \beta' \log(h_{t-1}^2) + \rho \left(\frac{\varepsilon_{t-1}}{h_{t-1}} \right)$$

where $\varepsilon_t \sim N(0, h_t^2)$, $D_1, \dots, D_6$ are dummies which take the value 1 in the i th day of the week and 0 in the others.⁸ The term δr_{t-1} accounts for mean reversion in price returns, the coefficients α' and ρ represent, respectively, the *size* and *sign* effects. Given the *inverse leverage effect*, the sign of ρ is expected to be 'positive', since *positive price shocks amplify the conditional price variance of the process more so than negative shocks*.⁹ Estimates, via maximum likelihood, are reported in Table 3, where reported estimates for ω' also account for the $(\sqrt{2/\pi})$ term. Looking at the mean equations of both models, it is possible to observe that the autoregressive coefficient is always significant and negative, confirming the expectations on processes reverting towards a long-term equilibrium level. Seasonality is also an important fact to account for. Moving to the conditional equation and comparing both performances, these models are persistent and stationary, since β and β' are significant and $\beta, \beta' \in (0, 1)$; finally, the *sign term* ρ , accounting for asymmetry, is always significant but, surprisingly, negative for all European markets apart from The Netherlands, where it is found to be positive. This provides the first empirical evidence that most European electricity spot markets exhibit a *normal* leverage effect, as documented formerly on the Italian PUN prices (see Petrella and Sapio, 2009).

As in Hadsell (2006), these results confirm that is very important to include the asymmetry into the model, and therefore an EGARCH(1,1) is more adequate than a simple GARCH(1,1) to detect electricity asymmetry.

⁷ EGARCH(1,1) was proposed by Nelson (1991) and implemented in electricity markets by Knittel and Roberts (2005).

⁸ The first dummy is for the return on Tuesday, the second dummy on Wednesday and so on, till Sunday which is dummy number six, the omitted day is Monday. Prices are referred to the day on which electricity is delivered.

⁹ From Knittel and Roberts (2005, p. 812).

Table 3: Estimates of ARCH-Type Models on the Entire Sample with p -Values in Brackets

	μ	δ	c_1	c_2	c_3	c_4	c_5	c_6	ω (or ω')	α (or α')	β (or β')	ρ
Germany												
M1	-0.287 (0.000)	-0.231 (0.000)	0.707 (0.000)	0.444 (0.000)	0.295 (0.000)	0.280 (0.000)	0.233 (0.000)	0.048 (0.002)	0.002 (0.000)	0.181 (0.000)	0.814 (0.000)	
M2	-0.283 (0.000)	-0.224 (0.000)	0.684 (0.000)	0.426 (0.000)	0.289 (0.000)	0.263 (0.000)	0.223 (0.000)	0.046 (0.001)	-0.413 (0.000)	0.335 (0.000)	0.947 (0.000)	-0.114 (0.000)
France												
M2	-0.274 (0.000)	-0.267 (0.000)	0.629 (0.000)	0.441 (0.000)	0.276 (0.000)	0.256 (0.000)	0.235 (0.000)	0.074 (0.000)	0.003 (0.000)	0.219 (0.000)	0.743 (0.000)	
M2	-0.262 (0.000)	-0.243 (0.000)	0.597 (0.000)	0.406 (0.000)	0.257 (0.000)	0.242 (0.000)	0.221 (0.000)	0.062 (0.000)	-0.528 (0.000)	0.353 (0.000)	0.918 (0.000)	-0.128 (0.000)
Spain												
M1	-0.222 (0.000)	-0.299 (0.000)	0.478 (0.000)	0.322 (0.000)	0.235 (0.000)	0.229 (0.000)	0.200 (0.000)	0.074 (0.000)	0.002 (0.000)	0.136 (0.000)	0.782 (0.000)	
M2	-0.227 (0.000)	-0.274 (0.000)	0.485 (0.000)	0.313 (0.000)	0.238 (0.000)	0.238 (0.000)	0.204 (0.000)	0.062 (0.000)	-0.443 (0.000)	0.226 (0.000)	0.928 (0.000)	-0.126 (0.000)
The Netherlands												
M1	-0.313 (0.000)	-0.335 (0.000)	0.733 (0.000)	0.499 (0.000)	0.333 (0.000)	0.307 (0.000)	0.265 (0.000)	0.064 (0.000)	0.004 (0.000)	0.291 (0.000)	0.727 (0.000)	
M2	-0.314 (0.000)	-0.329 (0.000)	0.755 (0.000)	0.500 (0.000)	0.349 (0.000)	0.313 (0.000)	0.266 (0.000)	0.079 (0.000)	-0.420 (0.000)	0.387 (0.000)	0.944 (0.000)	0.094 (0.000)
Italy												
M1	-0.202 (0.000)	-0.267 (0.000)	0.466 (0.000)	0.323 (0.000)	0.209 (0.000)	0.186 (0.000)	0.181 (0.000)	0.013 (0.119)	0.002 (0.000)	0.399 (0.000)	0.613 (0.000)	
M2	-0.203 (0.000)	-0.238 (0.000)	0.436 (0.000)	0.306 (0.000)	0.200 (0.000)	0.181 (0.000)	0.183 (0.000)	0.002 (0.785)	-0.713 (0.000)	0.408 (0.000)	0.892 (0.000)	-0.249 (0.000)

5. Sources of Electricity Spot Price Volatility

To identify possible causes for volatility in electricity markets, an hypothesis has been formulated to consider endogenous factors that affect market price mechanisms (and so market price variability), as prices of generation sources, regulatory mechanisms or market designs.

Among possible internal causes, the effects of virtual bidding (Hadsell and Shawky, 2007 on US volatility), market power (Mount, 2001), or again congestions and technologies that affect Italian zonal prices and volatilities (Gianfreda and Grossi, 2009) have been explored. Here, instead, volume effects on volatility¹⁰ have been investigated with several model specifications augmented by the daily average trading volumes, V_t , calculated for all markets with the logarithm of arithmetic means of 24 electricity quantities delivered in each individual hour, to be consistent with the specification of the conditional variance equations.

5.1. Basic and Augmented Models Proposed for the Normal Regime

The existence of a significant relationship between volatility and traded volumes is investigated proposing three models which are combinations of a basic model for the conditional mean (Equation 1) with three different specifications for the conditional variance, that is

- i) *SEGARCH* model, the exponential autoregressive conditional heteroskedasticity augmented by seasonality (Equation 4);
- ii) *VEGARCH* model, the exponential GARCH augmented by log-traded volume (Equation 5);
- iii) *SVEGARCH* model, which combines seasonality and volume in the variance equation (Equation 6).

Formally,

$$(4) \quad \log(h_t^2) = \omega' + \alpha' \left(\frac{|\varepsilon_{t-1}|}{h_{t-1}} - \sqrt{\frac{2}{\pi}} \right) + \beta' \log(h_{t-1}^2) + \rho \left(\frac{\varepsilon_{t-1}}{h_{t-1}} \right) + \sum_{i=1}^6 d_i D_i$$

$$(5) \quad \log(h_t^2) = \omega' + \alpha' \left(\frac{|\varepsilon_{t-1}|}{h_{t-1}} - \sqrt{\frac{2}{\pi}} \right) + \beta' \log(h_{t-1}^2) + \rho \left(\frac{\varepsilon_{t-1}}{h_{t-1}} \right) + \gamma \log \tilde{V}_{t-1}$$

¹⁰ Another choice could be to consider capacity, since when production approaches maximum available capacity then the price spikes. However this information is not available for all considered markets, so traded volume is used as a proxy for capacity.

$$(6) \quad \log(h_t^2) = \omega' + \alpha' \left(\frac{|\varepsilon_{t-1}|}{h_{t-1}} - \sqrt{\frac{2}{\pi}} \right) + \beta' \log(h_{t-1}^2) + \rho \left(\frac{\varepsilon_{t-1}}{h_{t-1}} \right) + \sum_{i=1}^6 d_i D_i + \gamma \log \tilde{V}_{t-1}.$$

These three models have been formulated and proposed to control for seasonal effects, and cross-effects between seasonality and traded volumes¹¹ also in the volatility dynamics. It is known that in power markets volume levels affect immediately prices according to available capacity, then the formulation with log-volume, consistent with the specification of the conditional variance equations, suggests a relation between *the magnitude of trading volume* and *the price changes*, as in Foster (1995) for crude oil futures and in Najand and Yung (1991) for equity markets. Moreover, given the market mechanisms of simultaneous determination of clearing market prices and quantities on spot markets, the lagged volumes are used to avoid problems of endogeneity.

Results have been summarized in Table 4,¹² where only the parameters' estimates for the conditional variance equation have been reported for all three model specifications and for all studied markets, when prices were cut at a threshold of 70€.

Looking at the estimates, it is possible to draw the following conclusions:¹³

- i) The *mean-reversion* property, in return dynamics, is confirmed in all markets apart for Italy. The δ coefficients of the autoregressive term are significant and negative, indicating that the previous return moves the following one down towards a long-term equilibrium level.
- ii) Seasonality is generally an important feature in both mean and variance equations, indicating that it also affects volatility dynamics (as in Hadsell *et al.* 2004).
- iii) Volatility clustering, persistence and asymmetry are confirmed as important features in modelling problems. In detail, being β' always significant, there is a clear indication that past electricity volatility increases future volatility values. Concerning the asymmetry, the

¹¹ The aim of these models lies in controlling if the inclusion of traded volume makes seasonality insignificant.

¹² Complete results are available on request.

¹³ This analysis, with its limitations on highly nonlinear cut filtering, represents a preliminary investigation on the existence of a possible volume–volatility relationship as in other financial markets and, consequently to results obtained, it will be interesting to apply more appropriate modelling to account also for results on the estimated relation coming from the *abnormal* regime and so considering the complete price dynamics. As suggested by a referee, the jump-diffusion and Markov switching literature could be considered, but this would represent a different analysis which could be addressed in future developments when controlling if changes in regimes are driven by traded volumes or other factors like temperatures as in Huisman (2008).

Table 4: Models Estimates for the Conditional Variance Equations with Price Spikes Set at the Threshold $J = 70\text{€}$

Markets	Models	Conditional variance equation										
		ω'	α'	β'	ρ	d_1	d_2	d_3	d_4	d_5	d_6	γ
Germany	SEGARCH	-1.720***	0.480***	0.868***	-0.193***	1.495***	0.503***	0.348***	0.758***	0.611***	1.046***	-0.035
	VEGARCH	-2.526***	0.644***	0.549***	-0.401***							-0.074**
	SVEGARCH	-3.389***	0.617***	0.448***	-0.444***	0.958***	0.667***	0.394**	1.124	0.153	0.708***	
France	SEGARCH	-0.682***	0.399***	0.921***	-0.210***	0.758***	-0.842***	-0.670***	0.067	-0.128	0.495***	-0.058***
	VEGARCH	-0.469***	0.316***	0.886***	-0.143***							0.122***
	SVEGARCH	-0.381***	0.456***	0.781***	-0.156***	0.233	-0.770***	-1.034***	-0.327**	-0.548***	-0.157	
Spain	SEGARCH	-1.099***	0.335***	0.920***	-0.196***	0.899***	0.253*	0.112	0.261*	0.310**	0.899	-0.002
	VEGARCH	-0.810***	0.394***	0.905***	0.019							-0.067***
	SVEGARCH	-1.232*	0.469***	0.855***	0.022	1.023*	0.408	0.344	0.303	0.509	1.087*	
Netherlands	SEGARCH	-0.802***	0.497***	0.926***	0.075***	0.557***	-0.363***	-0.391***	-0.010	0.282**	0.580***	0.023***
	VEGARCH	-0.443***	0.237***	0.963***	0.108***							0.039***
	SVEGARCH	-0.970***	0.406***	0.922***	0.081**	0.646***	-0.133	-0.183	0.089	0.141	0.403**	
Italy	SEGARCH	-6.059***	0.933***	0.134***	-0.400***	1.419***	-0.270***	-0.673***	-1.651***	-1.081***	1.390***	-0.454***
	VEGARCH	-1.440***	0.911***	0.367***	1.037***							-0.234***
	SVEGARCH	-3.764***	1.633***	0.113***	-0.874***	0.731	-2.357***	-1.508***	-2.518***	-2.616***		-0.234***

***, ** and * mean 'statistically significant at 1%, 5% and 10%, respectively.

sign term ρ , is expected and found always positive only in the Netherlands. Whereas it has a negative effect in Germany, France, in one specification for Spain and two for Italy. Therefore it is possible to conclude that only the Dutch market exhibits the *inverse leverage effect*, whereas German and French markets act as other financial markets characterized by simple leverage effect. Controversial results are found in Spain and in Italy, so it is possible to state that traders' responses to new information is somehow asymmetric.

- iv) Traded volumes effects are significant and negative,¹⁴ indicating that past volumes represent useful information which reduces next day volatility. On the other hand, the Netherlands exhibits a volatility which is amplified by traded volumes and indeed the sign of γ is positive, this could be due to the fact that the Netherlands always acts at margin without excess of capacity.¹⁵

6. Conclusions

After the deregulation processes observed in the electricity sector, producers and consumers, and in general all market participants, faced different types of risks and their management often involves derivatives in which price and volatility of electricity play an important role. Therefore, this paper has first contributed to the analysis of the volatility of five wholesale electricity spot markets of the Continental Europe. Secondly, significant volatility persistence and asymmetric effects have been documented, using GARCH models after filtering outliers. Interestingly only the Dutch market exhibits the stylized *inverse leverage effect*, whereas the other markets act as financial markets characterized by a simple leverage effect. Finally, a significant relation between volume and volatility has been proved on an empirical basis, finding that this link can be both positive or negative depending on the intrinsic structure of individual markets.

¹⁴ Interestingly, looking at France, the inclusion of volumes together with seasonality changes the signs of γ estimates. A reasonable explanation could be that some other forms of seasonality not detected are still affecting the dynamics.

¹⁵ In simple words, production plus import are almost always equal to demand plus export. Source data available from UCTE for years 2005–2008.

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Non-technical Summary

This paper analyses the volatility of five European wholesale markets for electricity. In economics and finance, volatility is the simplest and basic criterion to study risks of holding assets with uncertain future values. After the deregulation processes observed in the electricity sector, all market participants face different types of risks. As a consequence, energy derivatives often written on electricity have been formulated and traded, hence it has become important to monitor electricity prices and volatility. Because electricity is a commodity with certain special features it is important to understand its risk properties. This understanding can help, for instance, firms to formulate their supply schedules and evaluate real investments using modern asset pricing techniques; but also consumers to hedge against risk of sudden price variations or risks associated with physical delivery.

Electricity markets exhibit very high volatility, so it is essential to understand where it comes from. There is not clear evidence of price dynamics in response to demand variations in electricity markets. Previous studies tried to understand this relationship providing a not clear evidence. Therefore, this article contributes to the first investigation of European electricity spot volatility and provides empirical evidence on the volume–volatility relationships at spot price level, considering at the same time all features of electricity spot prices without excluding observations. Modelling electricity prices is a difficult task because of the existence of sudden upward usually followed by downward price movements. Given that it is usual to distinguish between two different regimes (a *normal* regime against an *abnormal* one, where the price jumps to high values reverting usually in short time to normal lower values), these spikes have been identified using threshold values, and so reduce the spiky magnitude which affects estimations. After filtering the anomalous values, hence focussing on the normal regime, the volatility of German, French, Spanish, Dutch and Italian electricity wholesale prices is examined and explored for its characteristics. Time series models have been tested and specified to find

an answer to the important question ‘*Where does electricity volatility come from?*’

To this aim, average daily prices and volumes have been computed and all features of these spot electricity prices have been shown. Therefore, the presence of a long-run tendency, seasonality, periods with high price movements and asymmetries have been investigated. Several models have been tested and a few statistical tests have been performed on return time series to identify the more appropriate models and verify the conjectures of different sign effects of new incoming information and volume impacts on price volatility.

Based on this empirical evidence provided in this article, it is possible to draw the following conclusions. First of all, European electricity spot prices confirm the tendency towards a long-term equilibrium level with seasonal movements in return and volatility dynamics. Secondly the volatility persistence is documented; that is there is a clear indication that past electricity volatility increases future volatility values. Concerning the asymmetry, controversial results are found across European markets and so it is possible to state that traders’ responses to new information depends on the market structure and it is somehow asymmetric.

Finally, to identify a possible cause for volatility in electricity markets, traded volumes have been considered and results show that they affect significantly, generally in a negative way, the spot volatility, and so indicate that past volumes represent useful information which reduces next day volatility. Only, the Netherlands exhibits a volatility which is amplified by traded volumes and this could be explained by the fact that this market always supplies electricity at the margin, that is without excess of generation capacity with respect to electricity demand.